

Chapter 15
Timer and PWM
Exercises

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Summary of Equations

- ▶ Timer clock frequency f_{CK_CNT} vs. CPU Clock Frequency f_{CK_PSC}

$$f_{CK_CNT} = \frac{f_{CK_PSC}}{PSC + 1}$$

- ▶ Timer frequency f_{Timer} with up-counting or down-counting mode:

$$f_{Timer} = \frac{f_{CK_CNT}}{ARR + 1}; (Timer\ Period = (ARR + 1) * Clock\ Period)$$

- ▶ Timer frequency f_{Timer} with center-aligned counting mode:

$$f_{Timer} = \frac{f_{CK_CNT}}{2 * ARR}; (Timer\ Period = (2 * ARR) * Clock\ Period)$$

- ▶ PWM duty cycle for Mode 1 (Low-True):

$$Duty\ Cycle = \frac{CCR}{ARR + 1}$$

- ▶ PWM duty cycle for Mode 2 (High-True):

$$Duty\ Cycle = 1 - \frac{CCR}{ARR + 1}$$

Calculating ARR

- ▶ Suppose Timer Clock Frequency = 80MHz
- ▶ Goal: Timer Frequency = 100Hz
- ▶ What should the ARR value be for 1. up-counting mode; 2. down-counting mode; 3. center-aligned counting mode?

Calculating ARR with prescaler

- ▶ Suppose CPU Clock Frequency = 80MHz, PSC=79
- ▶ Goal: Timer Frequency = 100Hz
- ▶ What should the ARR value be for 1. up-counting mode; 2. down-counting mode; 3. center-aligned counting mode?

Calculating the ARR

- ▶ Suppose a 16-bit timer has the following settings
 - ▶ CPU clock frequency is 4 MHz.
 - ▶ Prescaler PSC = 39
 - ▶ Counting direction: center-aligned counting
 - ▶ Desired timer frequency = 100 Hz
- ▶ Calculate the ARR.

Clock Frequency and Timer Frequency

- ▶ Suppose all registers are 16 bits. CPU clock frequency is 400 MHz.
- ▶ What is the maximum and minimum Clock Frequency, and Timer Frequency, assuming up-counting mode?

PWM duty cycle

- ▶ Suppose a 16-bit timer has the following settings
 - ▶ CPU Clock frequency is 16 MHz.
 - ▶ Prescaler PSC = 159
 - ▶ ARR = 1999
 - ▶ CCR = 499
 - ▶ Counting direction: up-counting
 - ▶ Output is set as PWM Mode 2 (High True).
- ▶ Calculate the timer frequency and PWM duty cycle.